ARTIN-REES CONTROL ON THE IMAGE OF EXT AGAINST I^N M

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- For a bounded above complex F of finite free A-modules the images of Hom_A(F, I^n M) -> Hom_A(F, M) are contained in I^{n-c} Hom_A(F, M) for n >= c, with c independent of n. invoked at proof lines 29601-29602. Supplies the uniform Artin-Rees exponent termwise.

Source reference. Original lemma 0G3K: sources/stacks-more-algebra/pinned-original-source.tex, source lines 29432–29432. Cited at proof lines 29601-29602 for the uniform containment of the images; the conclusion is used inside the counted span.

Lemma 0.2. *Let A be a Noetherian ring. Let $I \subset A$ be an ideal. Let $K \in D(A)$ be pseudo-coherent and let M be a finite A-module. For each $p \in \mathbf{Z}$ there exists an c such that the image of $\mathrm{Ext}^p_A(K, I^n M) \to \mathrm{Ext}^p_A(K, M)$ is contained in $I^{n-c}\,\mathrm{Ext}^p_A(K, M)$ for $n \geq c$.*

Proof. Choose a bounded above complex $P^\bullet$ of finite free A-modules representing K. Then $\mathrm{Ext}^p_A(K, M)$ is the cohomology of

$$\mathrm{Hom}_A(F^{-p+1}, M) \xrightarrow{a} \mathrm{Hom}_A(F^{-p}, M) \xrightarrow{b} \mathrm{Hom}_A(F^{-p-1}, M)$$

and $\mathrm{Ext}^p_A(K, I^n M)$ is computed by replacing these finite A-modules by I^n times themselves. Thus the result by Lemma 0G3K (and much more is true). □

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